Isometric Immersions of Negatively Curved Metrics

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Hilberts Theorem

Theorem (Hilbert, 1901)

There is no isometric immersion of a complete, smooth surface of constant negative curvature into $\mathbb{R}^3$.

Hilbert, D. (1901).Ueber Flächen von Constanter Gaussscher Krümmung. *Transactions of the American Mathematical Society*, *2*(1), 87–99. Retrieved March 1, 2026, from http://www.jstor.org/stable/1986308

Overview

1. Hyperbolic Geometry
2. Curvature
3. Codazzi Pairs
4. Hilberts Theorem
5. Extensions of Hilberts Theorem

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Section 1

Hyperbolic Geometry

Euclids Fifth Postulate (Euclid c. 300 BC)

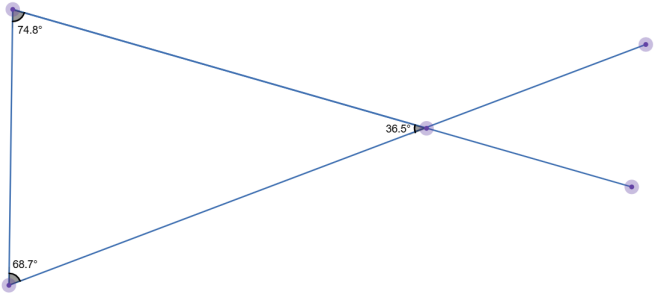
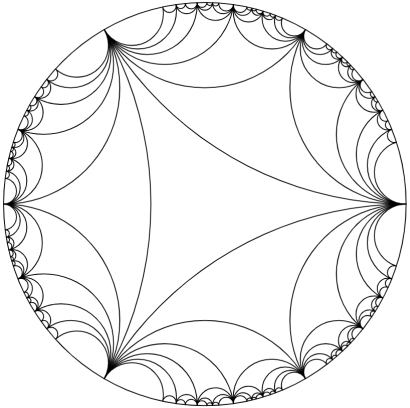


Figure: Euclid's 5th postulate

Poincaire Disc Model



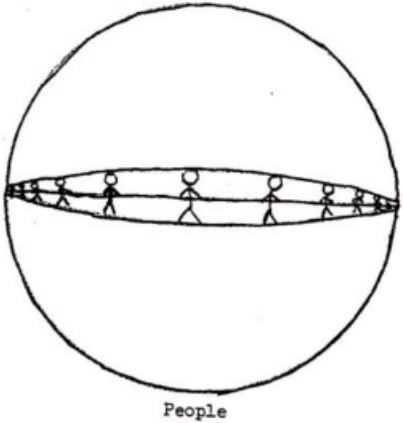


Figure: Poincaire Disc Model (Trott, 1999) + People on disc (Thurston, 1977)

Upper-Half Plane Model

- Let $\mathbb{H}^2 := \{z \in \mathbb{C} \mid Im(z) > 0\}$

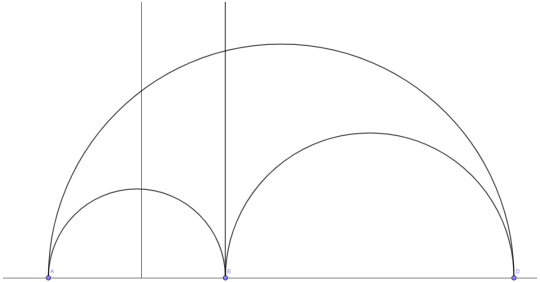


Figure: Upper-Half plane model

Isometries of the Hyperbolic Plane

- Isometries are given by elements of $PSL(2, \mathbb{R})$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot z = \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{R}, \;\; ad - bc = 1$$

Complete Hyperbolic Surfaces

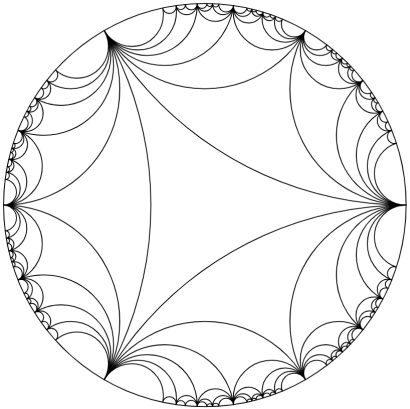
Definition

A (hyperbolic) surface is *complete* if its straight lines can be continued indefinitely.

Theorem (Hopf, 1926, Killing, 1891)

Any complete hyperbolic surface is the quotient $\mathbb{H}^2/\Gamma$ where Γ is a discrete subgroup of $PSL(2,\mathbb{R})$. These groups are called *Fuschian groups*.

Complete Hyperbolic Surfaces



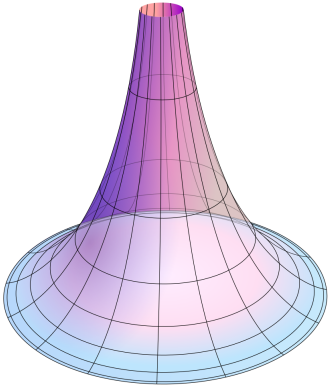


Figure: Poincaire Disc Model (Trott, 1999) and the Pseudo-sphere (WikiCommons)

Section 2

Curvature

Curvature

What is curvature?

Curvature

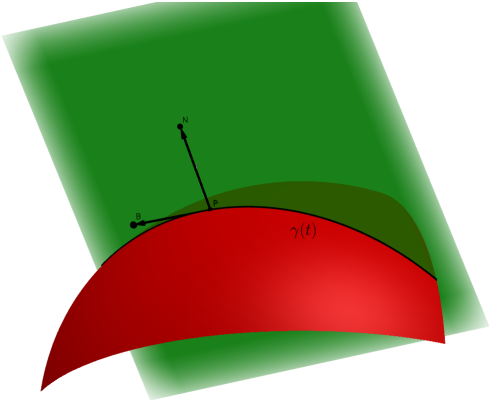


Figure: Normal section of a curved surface

Fundamental Pair

Definition

A *fundamental pair* on a manifold M is a pair (I, II) of symmetric, bilinear forms defined on TM such that I is a metric.

In coordinates,

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad II = \begin{pmatrix} e & f \\ f & g \end{pmatrix}$$

Definition

The shape operator S is defined by $I(SX, Y) = II(X, Y)$. In coordinates $S = I^{-1}II$

Curvature of a Fundamental Pair

Definition

The *extrinsic* curvature K of a fundamental pair (I, II), is defined to be

$$K = \frac{eg - f^2}{EG - F^2} = \kappa_1 \kappa_2 = \det(S)$$

where κ_1, κ_2 are the eigenvalues (principal curvatures) of S.

This is Intrinsic!

Theorem Egregium (Gauss 1827)

Let (I, II) be the canonical fundamental pair associated to a surface in $\mathbb{R}^3$. Then the quantity $K = \kappa_1 \kappa_2$ is intrinsic!

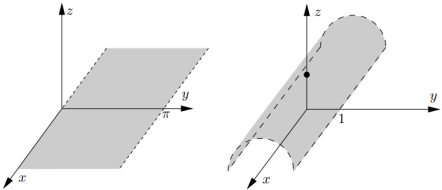


Figure: Two isometric surfaces with different principle curvatures (Lee)

What about surfaces not in $\mathbb{R}^3$?

Riemannian Manifolds and Isometric Immersion

Definition

A manifold M is Riemannian if there is an inner product $\langle \cdot, \cdot \rangle$ defined on each tangent space and $\langle \cdot, \cdot \rangle$ varies smoothly.

Definition

An isometric immersion

$$\varphi : (M, \langle \cdot, \cdot \rangle_1) \to (N, \langle \cdot, \cdot \rangle_2)$$

is a smooth, distance preserving map such that $d\varphi$ is injective everywhere.

Connections

Definition

Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold. A connection on the tangent bundle TM is a map

$$\nabla : \mathrm{Vect}(M) \times \mathrm{Vect}(M) \to \mathrm{Vect}(M)$$

that satisfies $\nabla_X fY = X(f)Y + f\nabla_X Y$ for $f \in C^\infty(M)$ and otherwise is linear.

Remark

There is a canonical connection associated to a Riemannian manifold called the Levi Civita connection. (Lee, 2018)

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Riemann Curvature Tensor

Definition

Let $(M, \langle \cdot, \cdot \rangle)$ be a Riemannian manifold. The Riemann curvature tensor is defined as

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z$$

Definition

The *sectional curvature* of $\Pi = \text{span}(e_1, e_2) \subset T_pM$ is defined as

$$sec(\Pi) = \langle R(e_1, e_2)e_1, e_2 \rangle$$

Space Forms

Definition

A *space form* is a complete Riemannian manifold M with constant sectional curvature.

Remark

If M is a complete simply connected Riemannian manifold with constant sectional curvature c

- $c = 0 \implies M \cong \mathbb{R}^n$
- $c = 1 \implies M \cong S^n$
- $c = -1 \implies M \cong \mathbb{H}^n$

Section 3

Codazzi Pairs

Gauss Equation

Let $\mathbb{M}^3(c)$ denote the unique simply connected space form with sectional curvature c and Σ be a surface.

Definition

The Gauss equation of an immersion $f : \Sigma \to \mathbb{M}^3(c)$

$$K(I) = K + c$$

where $K = \det(S)$ is the extrinsic curvature and $K(I)$ is the intrinsic curvature e.g. the sectional curvature $sec(T_p\Sigma)$.

Codazzi Equation

Definition

The Codazzi equation associated to the fundamental pair (I, II) with shape operator S is given by

$$\nabla_X SY - \nabla_Y SX = S[X, Y]$$

Codazzi Pair

Definition

A fundamental pair (I, II) with shape operator S is a Codazzi pair if the Codazzi equation is satisfied - i.e.

$$\nabla_X SY - \nabla_Y SX - S[X, Y] = 0$$

for $X, Y \in \mathrm{Vec}(\Sigma)$ and ∇ is the induced Levi-Civita connection on Σ.

Section 4

Hilberts Theorem

Asymptotic Curves

Definition

An asymptotic direction $v \in T_p\Sigma$ on Σ, is a direction where $II_p(v, v) = 0$. If $K < 0$ at $p \in \Sigma$ there are two asymptotic directions.

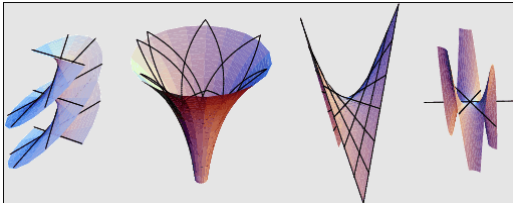

Figure: Asymptotic curves on some surfaces for the second fundamental form. (Wolfram)

Arc Length Paramaterization

Proposition (J. Gálvez, 2009)

Let (I, II) be a Codazzi pair on an oriented, simply connected surface of negative constant extrinsic curvature K. Then we can write (I, II) as

$$I = \begin{pmatrix} 1 & \cos\omega \\ \cos\omega & 1 \end{pmatrix}, \quad II = \begin{pmatrix} 0 & \sqrt{-K}\sin\omega \\ \sqrt{-K}\sin\omega & 0 \end{pmatrix}$$

with $0 < \omega < \pi$.

Change of Coordinates

Paramaterizing by curves (u, v) along the asymptotic directions

$$I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}, \quad II = \begin{pmatrix} 0 & f \\ f & 0 \end{pmatrix}, \quad f > 0$$

- If $\dfrac{\partial E}{\partial v} = \dfrac{\partial G}{\partial u} = 0$ then we can paramaterize by arc length as $E = E(u)$ and $G = G(v)$
- This will hold true if $\nabla_{\frac{\partial}{\partial u}} \frac{\partial}{\partial v} = \nabla_{\frac{\partial}{\partial v}} \frac{\partial}{\partial u} = 0$

Remark

Recall $\nabla_{\frac{\partial}{\partial u}} \frac{\partial}{\partial v} = \nabla_{\frac{\partial}{\partial v}} \frac{\partial}{\partial u} = \Gamma^1_{12} \frac{\partial}{\partial u} + \Gamma^2_{12} \frac{\partial}{\partial v}$

Can we parametrize nicely?

Letting $D = EG - F^2$ we can calculate the following conditions on the Christoffel symbols using Koszul's formula

$$\frac{D_u}{2D} = \Gamma^1_{11} + \Gamma^2_{12}, \quad \frac{D_v}{2D} = \Gamma^2_{22} + \Gamma^1_{12} \tag{1}$$

Now using the Codazzi equation we get the following condition on the Christoffel symbols

$$\frac{f_u}{f} = \Gamma^1_{11} - \Gamma^2_{12}, \quad \frac{f_v}{f} = \Gamma^2_{22} - \Gamma^1_{12} \tag{2}$$

Can we parametrize nicely?

Subtracting the corresponding equations of $(1) - (2)$ we obtain

$$\frac{f_u}{f} - \frac{D_u}{2D} = 2\Gamma_{12}^2, \quad \frac{D_v}{2D} - \frac{f_v}{f} = 2\Gamma_{12}^1 \tag{3}$$

We now have that $K = -\dfrac{f^2}{D}$ is a negative constant $\implies K_u = K_v = 0$.

Remark

These derivatives give exactly the numerator of the LHS and RHS respectively and so $\Gamma_{12}^2 = \Gamma_{12}^1 = 0$

Arc Length Paramaterization

Proposition (J. Gálvez, 2009)

Since $E(u,v) = E(u)$ and $G(u,v) = G(v)$ we can parametrize by arc length and write our metric as

$$I = \begin{pmatrix} 1 & \cos\omega \\ \cos\omega & 1 \end{pmatrix}, \quad II = \begin{pmatrix} 0 & \sqrt{-K}\sin\omega \\ \sqrt{-K}\sin\omega & 0 \end{pmatrix}$$

with $0 < \omega < \pi$ as $1 - F^2 > 0$.

Proof.

$$K = -\frac{f^2}{1-F^2} \implies F^2 + \left(\frac{f}{\sqrt{-K}}\right)^2 = 1$$

□

We have a global diffeomorphism!

Proposition (J. Gálvez, 2009)

The map $(u, v) : \Sigma \to \mathbb{R}^2$ is a global diffeomorphism.

Proof.

Let $III = du^2 - 2\cos\omega du dv + dv^2$. Then $I + III = 2(du^2 + dv^2)$ is a complete flat metric when I is complete and

$$(u, v) : (\Sigma, du^2 + dv^2) \to \mathbb{R}^2$$

is a local isometry, hence a covering map by completeness and hence a global diffeomorphism by simply connectedness. □

Hilberts Theorem

Theorem (Hilbert, 1901, J. Gálvez, 2009)

Let Σ be a complete surface of constant negative Gauss curvature $K(I)$. Then there is no isometric immersion

$$f : \Sigma \to \mathbb{M}^3(c)$$

for $c = 0, 1$ and if $c = -1$ there is no such immersion if $K(I) < -1$.

Hilberts Theorem Proof

Proof.

By our first theorem we can find global coordinates (u, v) defined in $\mathbb{R}^2$ such that

$$I = \begin{pmatrix} 1 & \cos\omega \\ \cos\omega & 1 \end{pmatrix}, \quad II = \begin{pmatrix} 0 & \sqrt{-K}\sin\omega \\ \sqrt{-K}\sin\omega & 0 \end{pmatrix}$$

and $0 < \omega < \pi$. The Gauss equation *requires* that

$$\frac{\partial}{\partial u}\omega_v = -(K + c)\sin\omega$$

where $-(K + c) > 0$. □

Section 5

Extensions of Hilberts Theorem

Non-constant negative curvature

Theorem (Efimov 1963)

Let Σ be a complete surface with Gauss curvature $K(I) \leq \delta < 0$. Then there is no isometric immersion

$$f : \Sigma \to \mathbb{R}^3$$

.

Strictly Seperated Curvatures

Definition

The principle curvatures, k_1, k_2, of a fundamental pair (I, II) are called *strictly separated* if there exists numbers c_1, c_2 such that

$$k_1 \leq c_1 < c_2 \leq k_2$$

Non-constant negative curvature

Theorem (J. A. Gálvez et al., 2015)

Let (I, II) be a Codazzi pair on Σ with strictly separated principal curvatures. If (Σ, I) is a complete surface with Gauss curvature $K(I) \leq 0$ and I is not flat then Σ is homeomorphic to a plane and

$$\int_\Sigma |K(I)| dA \leq 2\pi$$

where dA is the area form induced by I.

Non-constant negative curvature

Corollary (J. A. Gálvez et al., 2015)

Let $\varphi : \Sigma \to \mathbb{H}^3$ be an immersion with Gaussian curvature $K(I) \leq -1$, and one of its principle curvature functions satisfying

$$k_i \geq \epsilon > 0, \quad \text{for } \epsilon > 0$$

then φ is not a complete immersion.

Questions?

Questions?

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